

Empire Research Architecture: Game Design Specification

Version: 2.0
Status: Approved Architecture Spec
Scope: Core Progression, Backend Data Model, & Discipline Architecture

1. Executive Summary & Core Philosophy

The legacy research tree suffered from critical scaling limitations, forcing unrelated technologies into flat, arbitrary branches as the game expanded. The redesigned **Empire Research Architecture** transitions research from a single crowded tree into a *Forest Architecture* composed of five core Disciplines, each containing independent, specialized sub-fields of knowledge.

This design is anchored on four foundational principles:

- **Disciplines as Knowledge Collections:** Research is categorized into five macro disciplines (Economy, Military, Development, Intelligence, Society). Disciplines serve as structural containers rather than single linear tracks.
- **Multi-Tree Hierarchy:** Each discipline contains distinct sub-trees (Fields), allowing gameplay expansion without cluttering or misplacing technologies.
- **Data-Driven Milestone Loop:** Progression follows a **3-6-9-10 Milestone** system. Every milestone grants tangible gameplay rewards (new research nodes, specialized buildings, unit doctrines, or unique mechanics) rather than silent numerical percentage boosts.
- **Institutional Tier Progression (Tier II):** Tier II represents an era-defining shift governed by physical institutions (e.g., Royal Library, Imperial Academy, Officer Corps) that unlock entire new gameplay paradigms.

2. System Architecture: Forest vs. Tree Paradigm

In this architecture, players specialize deeply within specific fields rather than exhausting a shallow, generic branch. Below is the structural hierarchy across all five disciplines:

Discipline	Primary Sub-Fields	Core Strategic Focus
Economy	Agriculture, Lumber Industry, Mining Industry, Construction Industry	Resource extraction, stockpile protection, logistics capacity, and harvest efficiency.
Military	Tactics, Physical Training, Metallurgical Warfare, Ranged Ballistics, Horsemanship, Siege Warfare	Troop offense/defense, movement speed, tactical march size, and siege mechanics.
Development	Structural Engineering, Architecture, Metropolitan Planning, Municipal Logistics	City infrastructure, building speed/durability, population capacity, and resource field expansion.

Discipline	Primary Sub-Fields	Core Strategic Focus
Intelligence	Informatics, Reconnaissance, Counter-Espionage, Cartography, Signals	Map visibility, scout reporting detail, stealth mechanics, and anti-scouting countermeasures.
Society	Public Health, Education, Commercial Trade, Governance, Cultural Influence	Population growth, tax output, research speed efficiency, market rates, and diplomatic standing.

3. Progression & Milestone Unlocks (3-6-9-10 Model)

To prevent player fatigue caused by long research timers with minor linear stat increments, every technology features distinct milestone levels at 3, 6, 9, and 10.

Milestone Level	Unlock Type	Systemic Impact
Level 3	Branching Sub-Nodes	Unlocks secondary research branches within the field (e.g., Agriculture L3 unlocks Canals and Crop Rotation).
Level 6	Buildings & Unit Tiers	Unlocks specialized infrastructure (e.g., Granaries, Aqueducts) or advanced troop tier prerequisites.
Level 9	Doctrines & Mechanics	Unlocks active doctrines, battle formations, specialized march commands, or field passive traits.
Level 10	Mastery & Tier II Eligibility	Grants a unique field Mastery Perk and counts toward the discipline-wide completion requirement for Tier II.

Example Node Progression: Agriculture

- **Level 1:** Basic Farming (+5% Food Production per level)
- **Level 3 Milestone:** Unlocks *Irrigation* and *Crop Rotation* research sub-trees
- **Level 6 Milestone:** Unlocks *Grand Granary* city building construction
- **Level 9 Milestone:** Unlocks *Harvest Specialist* specialized troop doctrine
- **Level 10 Milestone:** Grants *Surplus Harvest* passive perk and satisfies Agriculture Tier II eligibility

4. Institutional Tier Progression (Tier II Era Shift)

Tier II transitions research from basic skill acquisition to institutional mastery. Tier II cannot be researched in a standard Academy; it requires building advanced research institutions once all Tier I technologies within a discipline reach Level 10.

Discipline	Tier I Scope	Tier II Institution	Tier II Field Name	New Gameplay Mechanics
Economy	Basic Gathering & Storage	Royal Library	Agricultural & Civil Engineering	Automated resource distribution, drought resistance, surplus harvest procs.
Military	Unit Stats & Basic Drills	Officer Academy	Imperial Officer Corps	Battle formations, command auras, veteran unit ascension, field marshals.
Development	Building Speed & Storage	Grand Academy	Urban Architecture	Metropolitan district expansion, instant building repair thresholds, material reduction.
Intelligence	Basic Scouts & Signals	State Cipher Bureau	Counter-Intelligence Bureau	Stealth scouts, detection of hidden reinforcements, false troop report generation.
Society	Tax & Population Boosts	Imperial Academy	Royal Academy of Sciences	Cross-empire tariff optimization, dynamic alliance passive buffs, stability locks.

5. Technical Implementation & Data-Driven Schema

The entire research system is engineered to be fully data-driven. Adding new fields, milestones, or balance tweaks requires zero backend code modifications—only updates to configuration data structures.

JSON Research Definition Schema

```
{
  "tech_code": "AGRICULTURE",
  "discipline": "ECONOMY",
  "field": "Agriculture",
  "max_level": 10,
  "base_time_seconds": 900,
  "base_cost": { "food": 500, "wood": 1000, "stone": 500, "iron": 200,
"gold": 0 },
  "req_academy_level": 1,
  "req_techs": {},

```

```

    "milestones": {
      "3": {
        "unlock_technologies": ["IRRIGATION", "CROP_ROTATION"],
        "description": "Unlocks Irrigation and Crop Rotation research
trees"
      },
      "6": {
        "unlock_buildings": ["BLD_GRAND_GRANARY"],
        "description": "Unlocks Grand Granary construction"
      },
      "9": {
        "unlock_trainings": ["TRN_HARVEST_SPECIALIST"],
        "description": "Unlocks Harvest Specialist doctrine"
      },
      "10": {
        "mastery_perk": "PERK_SURPLUS_HARVEST",
        "tier2_prereq_satisfied": true
      }
    }
  }
}

```

6. Backend Integration & Eligibility Validation

To integrate this specification into the engine, the system utilizes the following core formulas and logic check hooks:

Cost & Duration Formulas

- **Resource Scaling:** $\text{Cost} = \text{baseCost} * 1.45^{(\text{targetLevel} - 1)}$
- **Research Duration:** $\text{Duration} = (\text{baseTime} / (1 + (\text{MayorIntelligence} / 100))) * (0.95^{(\text{AcademyLevel} - 1)})$

Tier II Unlock Eligibility Algorithm

```

async function checkDisciplineTier2Eligibility(playerId,
disciplineName, db) {
  const disciplineTechCodes = Object.keys(TECH_DEFINITIONS).filter(
    code => TECH_DEFINITIONS[code].discipline === disciplineName
  );

  const res = await db.query(
    `SELECT tech_code, level FROM player_researches
    WHERE player_id = $1 AND tech_code = ANY($2::text[])`,
    [playerId, disciplineTechCodes]
  );

  const playerTechMap = new Map(res.rows.map(r => [r.tech_code,

```

```
r.level]]));

    for (const techCode of disciplineTechCodes) {
        const currentLevel = playerTechMap.get(techCode) || 0;
        if (currentLevel < 10) {
            return { eligible: false, reason: `Requires ${techCode} at Level
10.` };
        }
    }

    return { eligible: true };
}
```